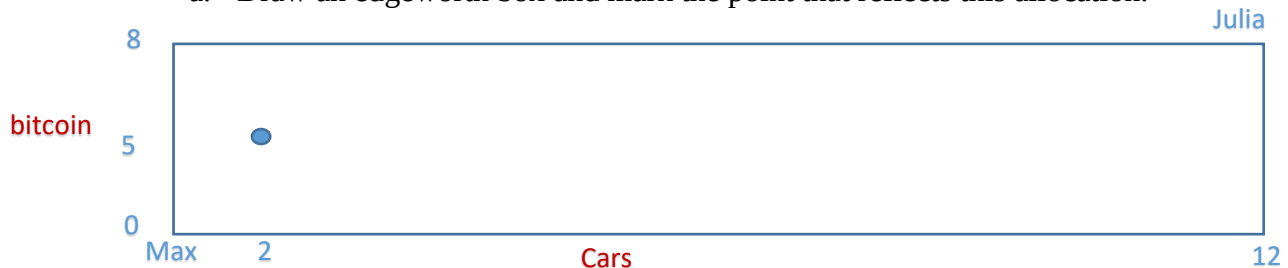


Problem set 2: Necessary conditions for the functioning of markets

Indifference curves and the Edgeworth box

1. Explain what is an indifference curve and we usually assume that it is convex with respect to the origin. *An indifference curve is the set of combination of two goods where the consumer's utility remains constant.*
2. Explain what is the “marginal rate of substitution” *It's how much of one good the consumer is willing to give up to get another good. Also – the ratio of prices and the total derivative of the indifference curve.*
3. Draw an indifference curve and a budget line. In your drawing, mark the point in which the utility is maximal and explain why that point is the relevant one. *See the slides. The relevant point is where the curve is tangent to the budget line.*
4. Suppose that Max owns 5 bitcoins and 2 cars, whereas Julia owns 3 bitcoins and 10 cars.
 - a. Draw an edgeworth box and mark the point that reflects this allocation.



- b. Assume that both Max and Julia have a standard indifference curve. Draw these curves and mark the area that reflects Pareto efficiency. *This is identical to the slides – the area of Pareto efficiency is the yellow area between the curves.*
5. Explain what is the “contract curve” and why it reflects Pareto efficient allocations. *The contract curve is the set of points where indifference curves to two consumers in the edgeworth box are tangent. These are points where both are willing to be in and there is no room for improvement, hence it is Pareto efficient.*

Necessary conditions for trade

6. What are the two important features of a competitive equilibrium? *Market clearing and price taking.*
7. What is the first theorem of welfare economics? What is the second theorem of welfare economics? Explain what the second theorem adds to our understanding of allocative efficiency. *See the slides. The first theorem says that a competitive equilibrium is efficient. The second theorem adds that it doesn't matter how we allocate –the market will yield an efficient outcome, so that we can reallocate and maintain efficiency.*

8. The government is thinking about taking some fixed amount of money from person A and giving it to person B. Suppose that the market is fully competitive. Person A is suing the government, claiming that this is “very inefficient”. Explain what that claim may be incorrect based on the second theorem of welfare economics.

The 2nd theorem says it doesn't matter how we allocate, hence if the government just transfers money from A to B (reallocation) that will not affect efficiency.